

Flying Straight

Pupil sheet

Name: _____

Date: _____

1. What I see on the screen

On the screen you see two numbers — **Roll** (tilt to one side) and **Pitch** (tilt forwards and back) — and a **circle** that turns **green** when both are below 3° .

The challenge: keep the circle green for **ten seconds**. If you leave the green, **the timer starts again from zero**.

Three rules. The box rests **on the desk**, not in your hand — no hand holds three degrees. **Nobody touches the table** during an attempt. And **the cable lies slack**, because a taut cable pulls the box crooked.

2. What I do on my own

First I predict. I do not change my prediction after I measure.

I put them in order, from the most level to the most tilted — *what I think*:

1. _____ 2. _____ 3. _____ 4. _____

I think so because _____

Now I measure. I put the box **in the middle** of the surface, with the same edge towards me every time, and take my hands off it.

Surface	Roll ($^\circ$)	Pitch ($^\circ$)	Does it go green?
My desk			yes / no
The window sill			yes / no
The floor			yes / no
With a book underneath			yes / no

The real order, after measuring:

1. _____ 2. _____ 3. _____ 4. _____

The challenge. The number of attempts is not an embarrassment — what counts is what I changed between them.

Attempt	I held it (s)	What I changed before the next one
1		
2		
3		
4		

3. The conclusion I write

I predicted that _____

The data show that _____

A pilot who can see nothing through the window relies on _____

Exit ticket

What did you change between attempts?
